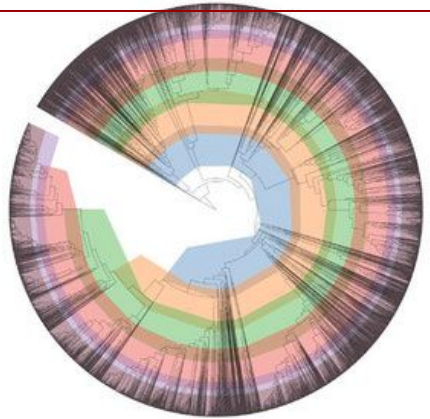
ENVIRONMENTAL METAGENOMICS

Physalia course, online, 12-16 October 2026

Long read assembly

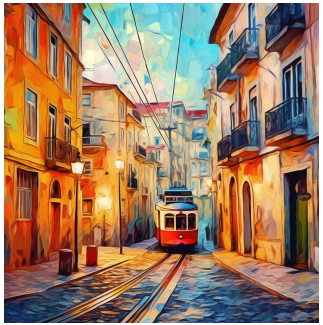
Nikolay Oskolkov, Lund University, NBIS SciLifeLab
Luis Pedro Coelho, Queensland University of Technology



Physalia Courses

NB: original course material courtesy:
Dr. Antti Karkman, University of Helsinki
Dr. Igor Pessi, Finnish Environment Institute (SYKE)

A bit about me



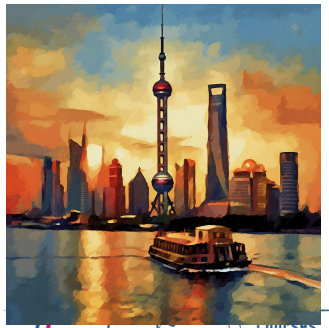
U. Lisbon
MS in computer science



Carnegie Mellon U
PhD in computational biology



EMBL (Heidelberg)
Postdoc (Bork group)



Fudan University
Junior PI



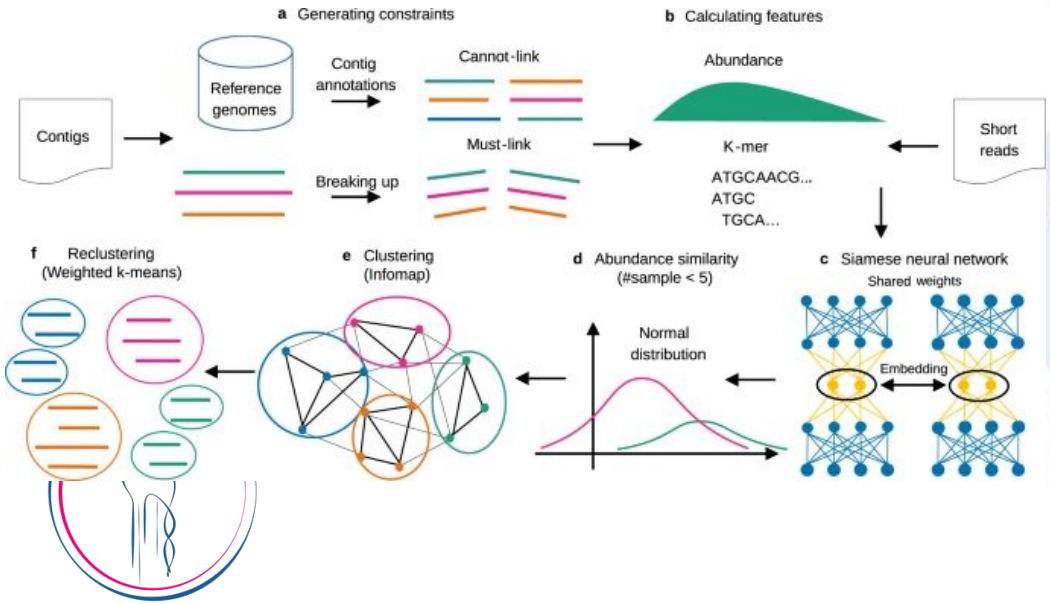
QUT
Associate Professor

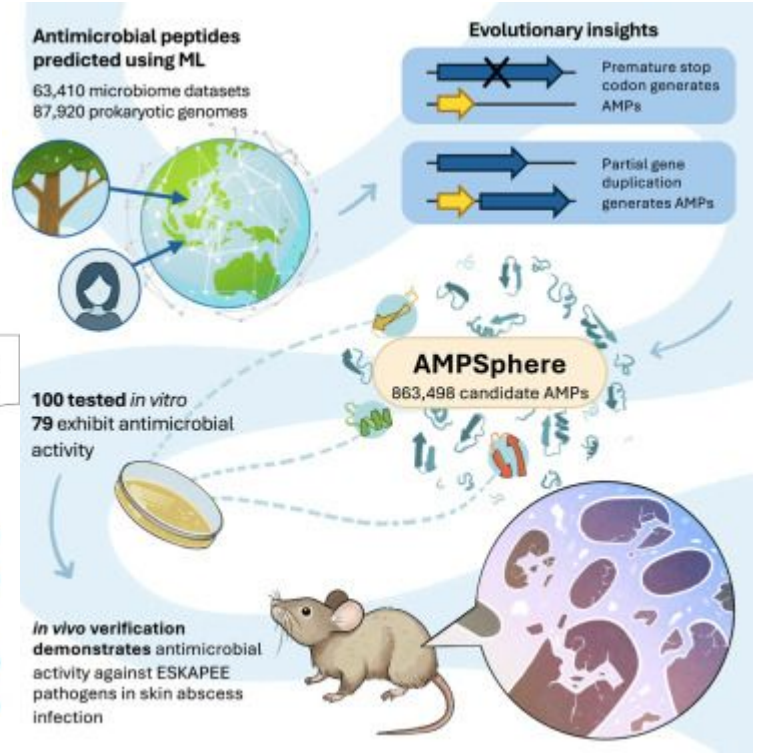


Today, I'm remote

A bit about my work

Combines method development with application-driven projects





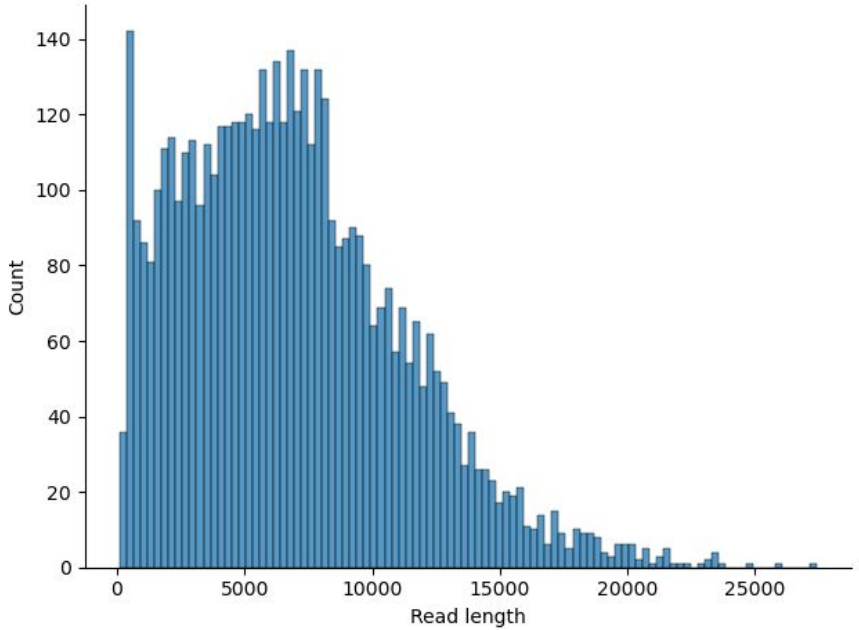
Long reads are longer than short reads

When short reads first appeared, they were 35bps!

Nowadays, short reads are 150-300bp and long reads are longer



PacBio



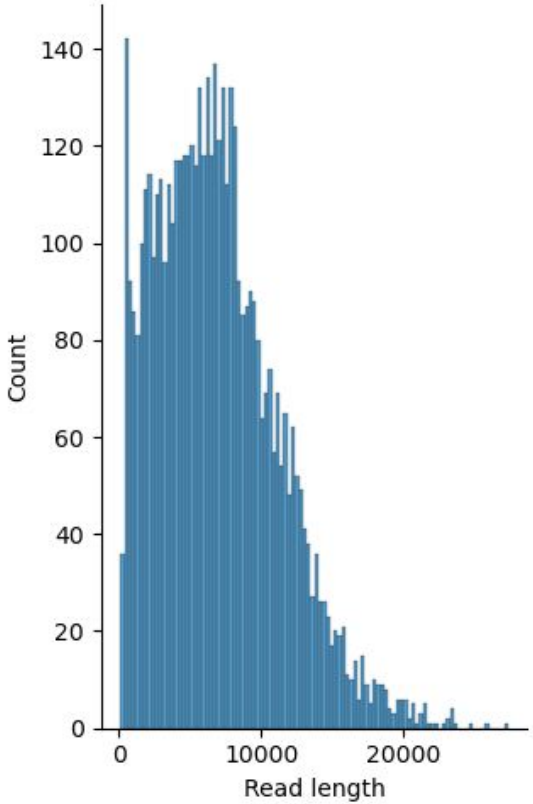
We can get better assemblies with long reads!

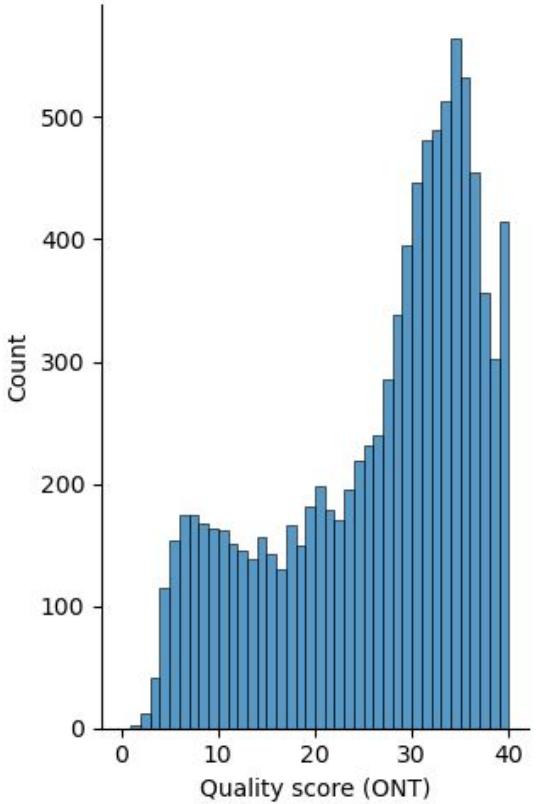
Some individual reads are longer than contigs from short-read assembly

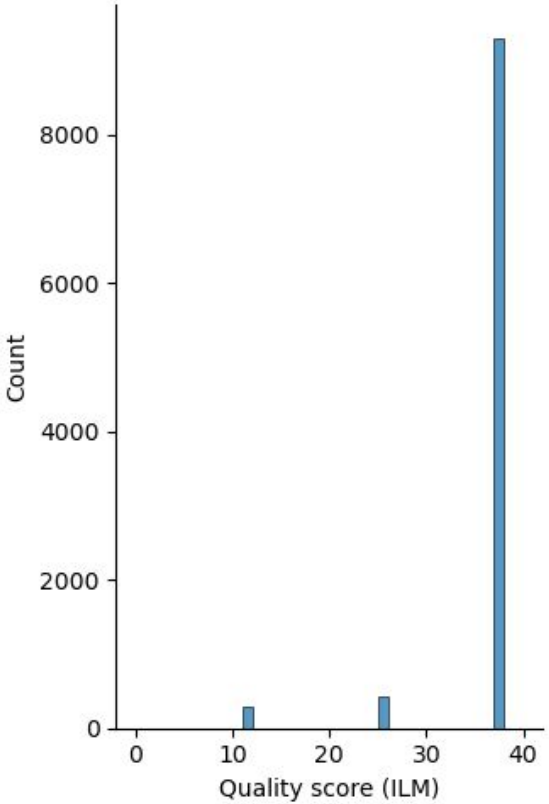
It can cover repeats and other hard to assemble regions!



Why do we even use short reads then?





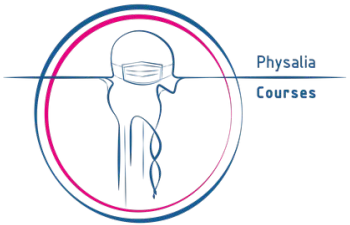


Best of both worlds: do both!

In fact, the data I showed before was from hybrid assembly!

How do you handle hybrid assembly?

1. Polish the long-reads with Illumina
2. Use a hybrid assembler (that takes ONT + ILM)
3. Polish the output of a long-read assembler with ILM
4. Use a short read assembler, then scaffold with the long reads
5. …?

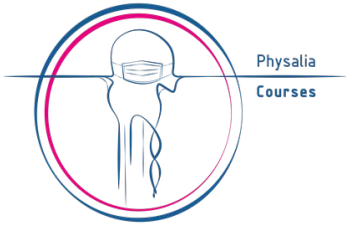


ONT data is in FASTQ format (like Illumina)

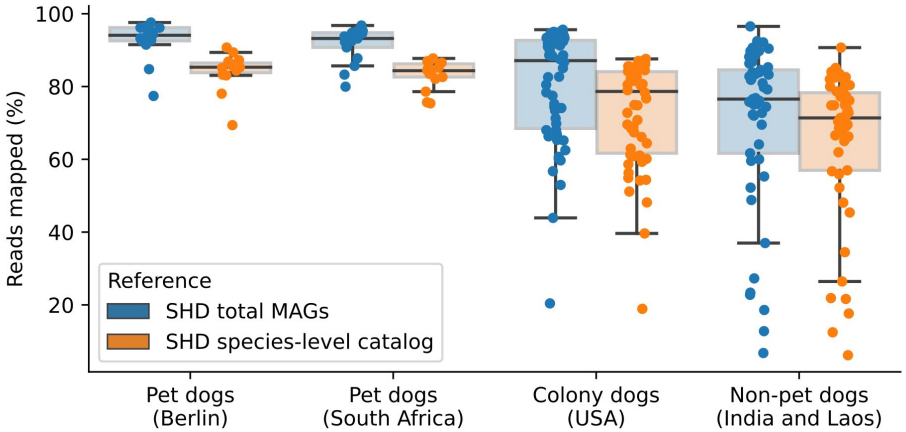
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+
+./:<;::>><@><??AEA7436<=F{GEA<=;=<;;<@????@AA?@BD=889:ACCA>=>@AABBD@@??BDJEEFFEBDCELFAA@@@@C;;;:DCDECCDC778?=CHHGECDDDFBAAAFAAA?=<<=?AADEF=;;:===B
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+
'+*+++322;>?B@<:==;;44('*,-/----,,,,==@ADC<;;9***+BED@=<<?>;DC>>=CD3333BAEB@666>CAABECB@AAA@BBAECB@?>?---1..217666?B@EGCCC?B???D>EIFDDLFD@@@ACDDFBB
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```

How do results look like with long-reads?

		#contig	Average length	N50
Sample A	mNGS	277,884	1,865	8,778
	PacBio-HiFi	2,575	99,842	269,406
	Nanopore	3,414	65,685	199,639
Sample B	mNGS	146,857	2,402	28,310
	PacBio-HiFi	1,447	157,228	1,081,788
	Nanopore	1,795	123,512	658,841
Sample C	mNGS	170,813	1,956	18,485
	PacBio-HiFi	822	171,215	1,270,126
	Nanopore	1,370	126,533	891,411

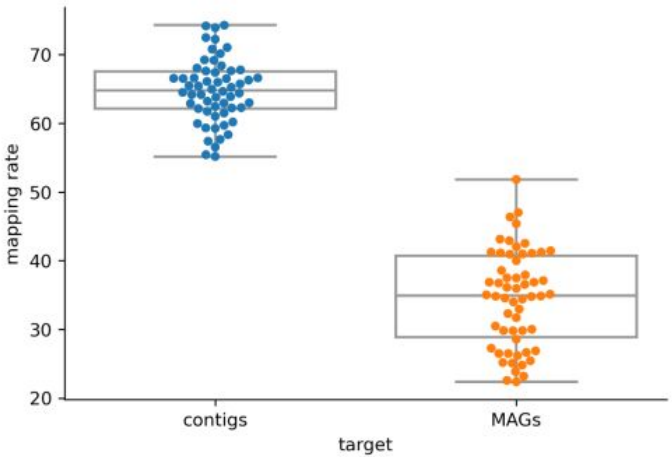


Results depend on the habitat!



Dog gut

(20 Gbp + 20 Gbp) x 50 samples



Soil

(40 Gbp + 40 Gbp) x 52 samples

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In dog guts, 20 Gbp was enough to get excellent representation of the community, but in soil even double that was not enough! — could also be Euks!

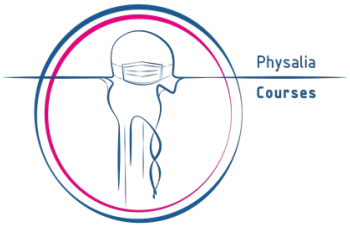
From Anna Cuscó (dog) & Yiqian Duan (soil)

A few other notes on long-reads

- Still a very rapidly evolving field (both in the wetlab & in the drylab)
- Differences between ONT versions matter a lot
- We might be able to get even more information in the future (methylation?)
- **Long-reads are the future!**

Tools are evolving quickly too

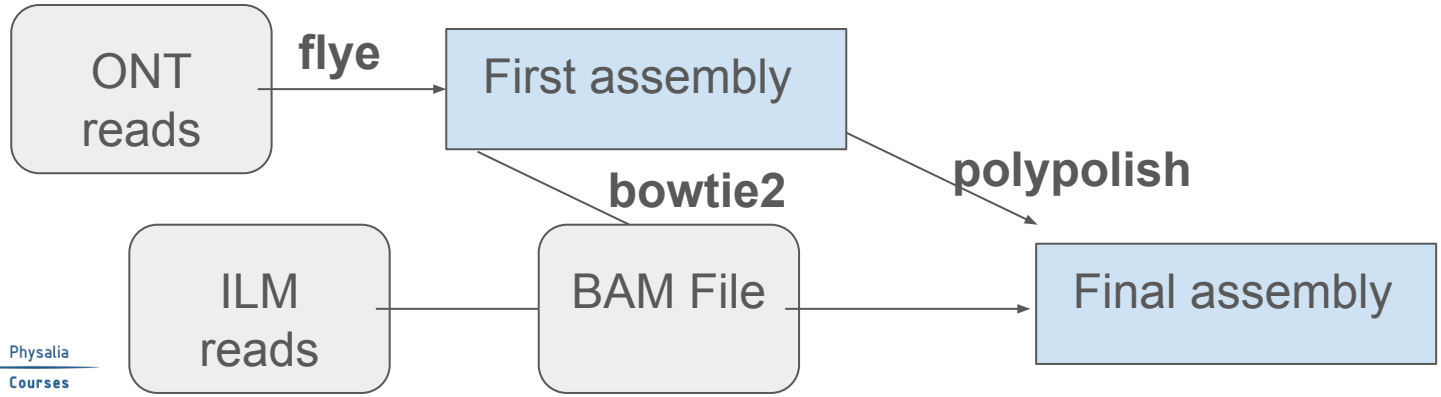
- We will show (meta)flye, but there are other tools, like myloasm (2026)



Exercises overview

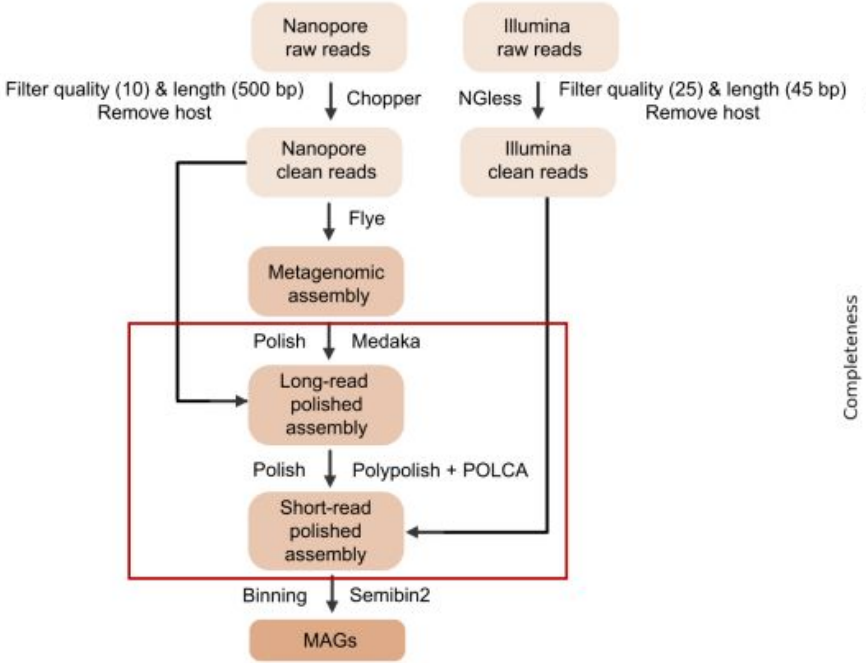
1. Assemble with **flye** (long-read only)
2. Polish with **polypolish** (using the short reads)

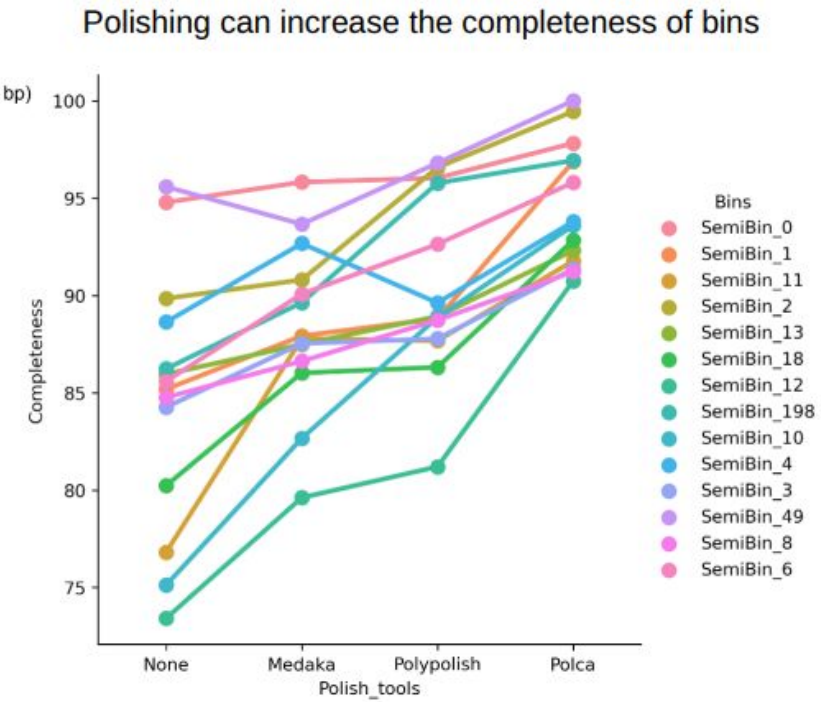
In order to polish, we need to align the short reads to the contigs from the **flye** assembly



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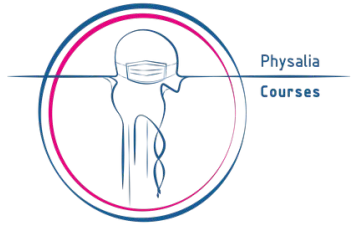
Real workflow

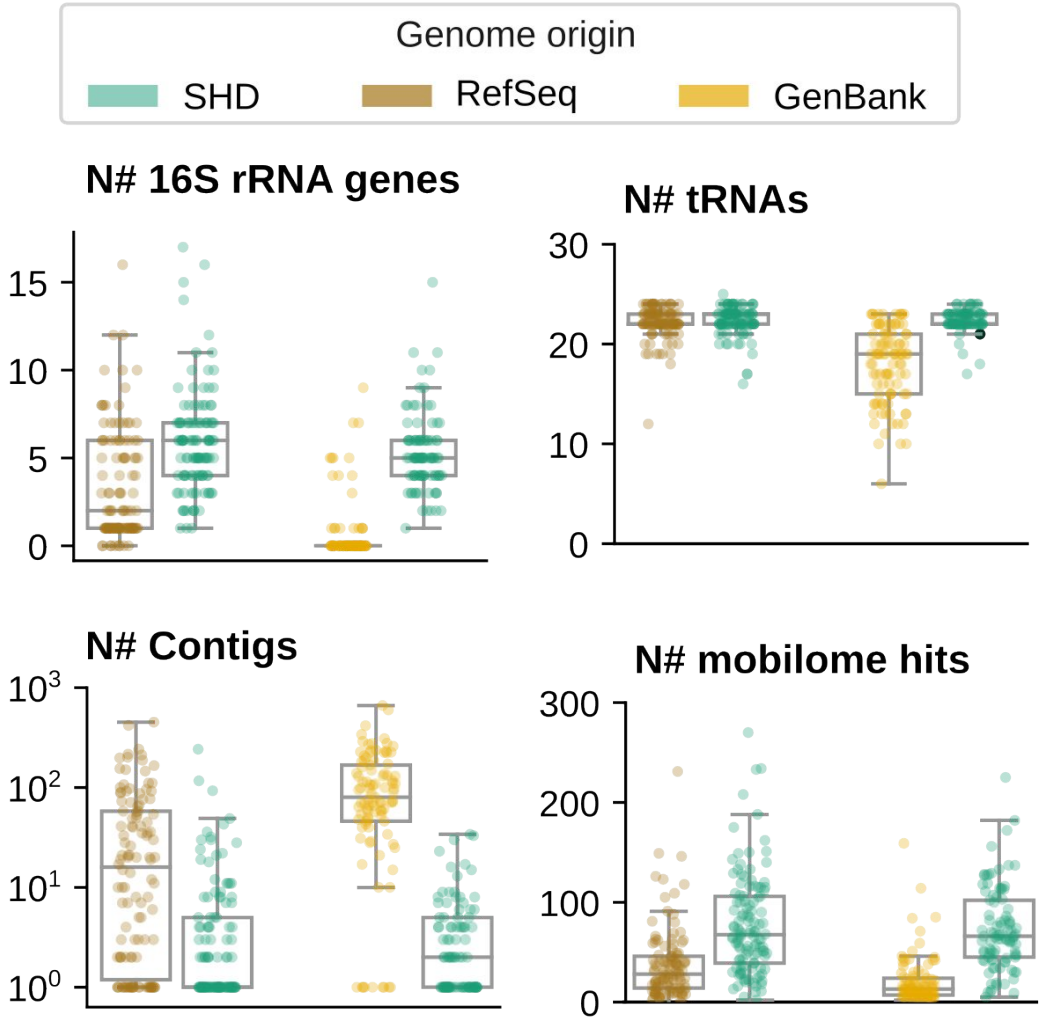




We can get MAGs that are better than the reference genome for that species!

Because the reference comes from an isolate, but was done with short reads.



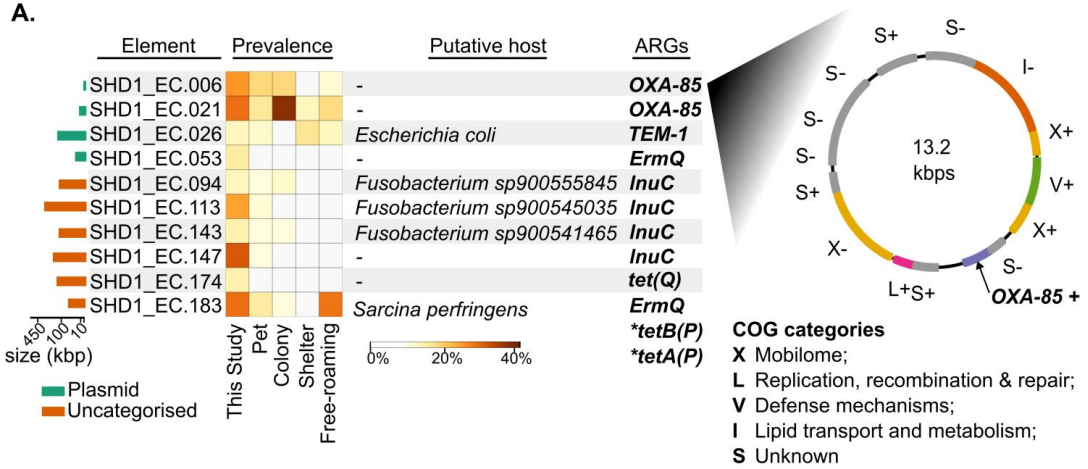


With long reads, we get extra-chromosomal elements

Capturing global pet dog gut microbial diversity and hundreds of near-finished bacterial genomes by using long-read metagenomics in a Shanghai cohort

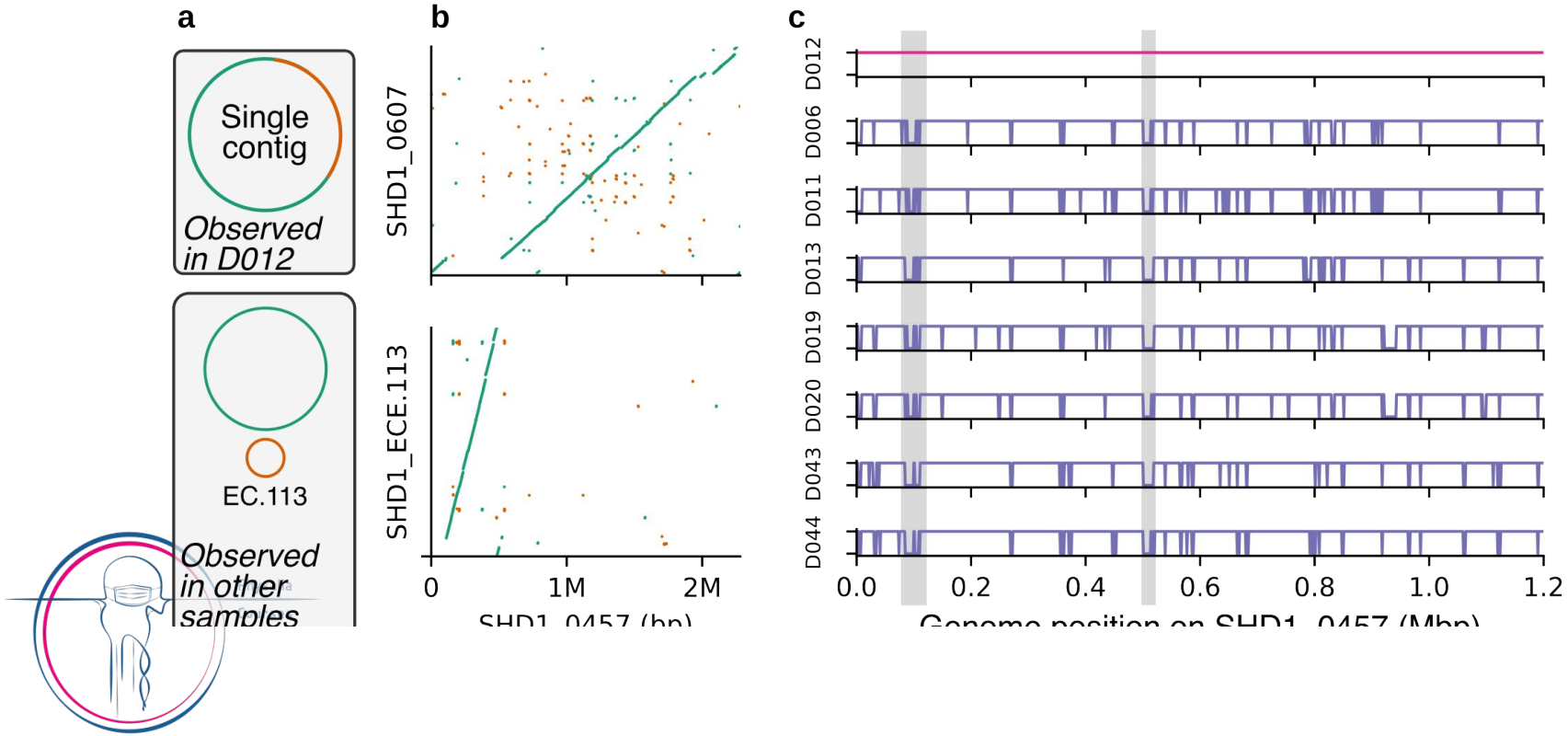
Anna Cuscó, Yiqian Duan, Fernando Gil, Alexei Chklovski, Nithya Kruthi, Shaojun Pan, Sofia Forslund, Susanne Lau, Ulrike Löber, Xing-Ming Zhao, Luis Pedro Coelho

doi: https://doi.org/10.1101/2025.09.17.676595



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We can also observe structural re-arrangements



Data that we will work on: urban soil microbiome

Long-read metagenomic sequencing reveals novel lineages and functional diversity in urban soil microbiome

Yiqian Duan, Anna Cuscó, Yaozhong Zhang, Juan S. Inda-Díaz, Chengkai Zhu, Alexandre Areias Castro, Xinrun Yang, Jiabao Yu, Gaofei Jiang, Xing-Ming Zhao, Luis Pedro Coelho

doi: https://doi.org/10.64898/2026.03.20.713087

We will use a toy version!

Not for real analyses!

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